

MERLIN SUSAN MATHEW

AI/ML ENGINEER

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https://github.com/Merlinsusan Kerala, India



EXPERIENCE

Graduate apprentice

C-dac ministry of electronics and information technology, government of india

- 2024 - 2025 Kerala, India
- Developed and deployed YOLO-based object detection models to automate traffic light control in coal mines, reducing collisions between heavy and light vehicles.
- Integrated IP and thermal cameras and obtained real-time video streams using the RTSP protocol for AI-based monitoring.
- Designed PyTorch and OpenCV-based models for classification and recognition tasks.
- Conducted image preprocessing (noise reduction, resizing, normalization) to enhance detection accuracy.
- Developed an interactive React.js dashboard for real-time data visualization and management.

EDUCATION

B.tech electronics and communication engineering

Apj abdul kalam technological university

2020 - 2024 Kerala, India

PROJECTS

Intelligent Q&A Chatbot with FAISS and Open AI LLM Integration

08/2025 Kerala, India

Developed a Question-Answer Chatbot using FAISS, Open AI, and large language models (LLMs) to efficiently retrieve and answer queries from uploaded documents and datasets.

- Achieved 88% accuracy in providing correct and context-aware answers, improving information retrieval speed and user experience.

Intelligent PDF and Video Summarizer Using NLP

06/2025 Kerala, India

Developed a PDF and Video Summarizer using NLP and computer vision techniques to automatically extract key insights and generate concise summaries from documents and video content.

- Enabled quick content review by reducing reading and viewing time while maintaining key information accuracy.

Early Diabetes Detection Using ML Algorithms

03/2025 Kerala, India

Built a Diabetes Prediction model using machine learning model to predict diabetes using key medical features.

- Achieved 92% accuracy and 0.89 ROC-AUC for reliable early detection of diabetes.

AI-Based Traffic Control System

08/2024 Kerala, India

Developed a YOLO-based object detection model for real-time vehicle detection as part of an AI-based traffic control system project for haul road crossings in coal mines.

- Achieved 95% accuracy in detecting heavy vehicles, enabling real-time traffic monitoring, improved safety, and enhanced operational efficiency at haul road crossings.

SUMMARY

Electronics Engineer with 1 year of experience in developing machine learning, deep learning, large language models (LLMs), and AI automation solutions. Skilled in YOLO object detection, RAG pipelines, LangChain, AI agents, and embedded AI hardware including Jetson Orin Nano. Experienced in solving real-world problems using data-driven approaches, automation, and advanced AI technologies.

LANGUAGES

English

Proficient



Malayalam

Native



SKILLS

RAG	LLM training	CUDA		
LLM fine-tuning		embeddings	OpenAI	
Deep Learning		Git	GitHub	LangChain
NLP	NLTK	NumPy	OpenCV	AWS
Pandas	Python	Machine Learning		
RNN	LSTM	PyTorch	Sagemaker	
sklearn	SpaCy	SQL	TensorFlow	
Zapier	PyTorch	Microsoft Word		n8n
Data Analysis		Vector DB	Docker	
MLOps	Open Ai	FAISS		